

RMIT University

OENG118X/COSC250X Engineering & IT Capstone Project



Capstone Project Final Report

Offline Video Analysis

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1 Executive summary

The project, Traffic Offline Video Analysis, aims to address the growing challenge of monitoring and enforcing traffic regulations, particularly in urban areas with high vehicle volumes. As traffic violations like red light crossing and illegal parking become more common, law enforcement agencies face difficulties manually identifying and penalizing offenders. This not only affects traffic flow but also compromises road safety. The need for an efficient, automated solution to detect these violations has become evident.

The core objective of this project is to develop a system capable of analysing offline traffic footage and identifying violations such as red light crossing and illegal parking. By leveraging advanced artificial intelligence (AI) models, the system will automate the detection process, significantly reducing the need for human intervention in video analysis. The AI models will be trained to accurately recognize these specific traffic violations, enabling quicker and more reliable enforcement.

The project consists of two main components: an AI-based video analysis engine and a user-friendly front-end website. The AI models will be responsible for detecting violations from uploaded video footage, while the website will allow users to easily input videos for analysis and review the results. This dual structure ensures both technical functionality and accessibility for end users, such as traffic authorities and law enforcement personnel.

The significance of this project extends beyond its practical applications in traffic law enforcement. From a theoretical perspective, it contributes to the growing field of AI-based video analysis and computer vision, showcasing how machine learning models can be applied to real-world problems. Practically, it has the potential to enhance road safety, reduce human error in traffic violation detection, and improve the efficiency of law enforcement processes. By providing a scalable solution to an everyday issue, this project offers a novel approach to traffic regulation that could be adopted in cities worldwide.

2 Project background & problem statement

The widespread installation of surveillance cameras and sensor networks in cities has generated a vast amount of data that can enhance traffic management and safety. However, the sheer scale and complexity of this data present major challenges for manual processing and interpretation. Current automated systems for identifying traffic violations often suffer from high false-positive rates, limited adaptability to changing traffic conditions, and difficulties with scaling. Real-time traffic detection options are available, however, with the high cost and resources needed to maintain it, it is very hard and costly to install this system in every traffic choke point all over the country. Consequently, authorities struggle to enforce traffic regulations effectively and address emerging safety issues efficiently.

Considering these challenges, our research aims to develop an innovative machine-learning system for detecting traffic infractions. By analysing video data from security cameras, we intend to identify violations such as running red lights and illegal parking using computer vision algorithms and deep learning techniques. Our approach involves training robust models on large-scale datasets, enabling accurate and reliable detection of infractions across various environmental and traffic conditions.

Traffic violations pose significant threats to public safety and urban mobility. Despite authorities' rigorous efforts to enforce traffic laws, traditional monitoring and detection systems have failed to keep pace with the complexity and scale of modern road networks. Manual enforcement is labor-intensive, prone to human error, and often cannot adapt to the dynamic nature of traffic patterns and behaviors.

Our project proposed a solution for transforming the way that traffic violations are detected, handled, and enforced. We intend to address the following important difficulties using machine learning and computer vision:

1. **Improve Traffic Management:** Automated violation identification simplifies enforcement operations, allowing authorities to better deploy resources and focus on high-risk locations like intersections and school areas,

where infractions are most common. This can be done by the analysis of traffic volume as well as the moving trends of vehicles.

2. **Improve Road Safety:** Accurate and fast identification of traffic offenses allows authorities to respond proactively and reduce the likelihood of accidents and injuries on the road, saving lives and relieving the pressure on emergency response services.

3. **Data-Driven System:** Our machine learning algorithms' insights give important knowledge to politicians and urban planners, allowing for evidence-based actions to address systemic issues influencing road safety and traffic flow.

4. **Scalability:** Our system is intended to be scalable and adaptable to a variety of urban surroundings and traffic circumstances, assuring its efficacy across several geographic areas and transportation networks.

3 Literature review & market research

The final phase of the project requires extensive development as well as completion of every component in the entire application pipeline. Therefore, a detailed description of the complete flow is necessary to identify the key features of the analyser. As established at the beginning, the full pipeline is composed of two fundamental sections: the webpage along with the AI-integrated analyser in the backend. Since the main emphasis of the last phase was the Automatic License Plate Recognition (ALPR) unit, the focus is shifted to the web interface, violation detection models, and databases as well as the incorporation of those modules at this stage of the project. Hence, our team can effectively select relevant materials for research. This section of the report provides the findings we have synthesized from various scientific sources that are beneficial to our development of the components. In addition to that, we will also document how our proposed structure contributes to the expansion of traffic video analysis with AI in terms of both practical and theoretical purposes.

The majority of the literature studied during this final phase revolves around how AI models are utilized for detecting traffic violations. Our project group and the company supervisors have agreed upon the 2 types of violations to inspect in the traffic videos: red light and parking violations. Regarding red light violations, in particular, surveillance videos at intersections or any location equipped with a traffic light system will be recorded. Our analyser is responsible for detecting, counting, and recording the plate numbers of vehicles failing to comply with the red-light signal. A similar methodology is implemented concerning parking violations with the extra task of detecting road signs to distinguish between no-parking and no-stopping instructions.

Zone-based vehicle counting

The first paper "Object Tracking and Counting in a Zone using YOLOv4, DeepSORT, and TensorFlow" by Kumar et al. presents an approach for tracking and counting multiple objects in a specific zone using a combination of the YOLOv4 object detection algorithm and DeepSORT for tracking [1]. The proposed methodology entails three modules: object detection, tracking, together with zonal counting. Firstly, YOLOv4 is employed to detect objects in each frame of the video. YOLOv4's advantage lies in its ability to detect multiple objects in real-time by applying a single convolutional neural network (CNN) to predict bounding boxes as well as class probabilities. Furthermore, pre-trained weights from the COCO dataset are used, which contain 80 classes of commonly found objects, including vehicles and people. Next, DeepSORT is integrated with YOLOv4 for object tracking. DeepSORT extends SORT by attaching an appearance feature vector to each detected object, facilitating object tracking with greater accuracy. The appearance vector is generated using a convolutional neural network (CNN), which extracts features that distinguish different objects even in crowded scenes. Kalman filters are also used in this process to predict the motion of objects, making the algorithm robust against occlusion, i.e., when one object is blocked by another. The final phase is zonal counting in which a specific zone is defined within the video frame, and objects entering or exiting this zone are counted. With the help of OpenCV, the authors construct the zone in the video feed, changing its color based on the number of objects within it. Should the object count exceed a predefined threshold, an alarm is triggered, making this system applicable for crowd control during events like the COVID-19 pandemic, where monitoring group size is crucial. The system is evaluated on the MOT17 dataset, a standard dataset used for benchmarking multiple object-tracking algorithms. MOT17 consists of real-world video sequences captured under various conditions, including scenes with moving cameras and crowded pedestrian areas. The authors use the MOTA (Multiple Object Tracking Accuracy) metric to evaluate their model's performance, which accounts for false positives, false negatives, and identity

switches. The system achieved a MOTA score of around 60% across different test sequences, with a tracking speed of up to 31 FPS on a Jetson Xavier NX. Apart from that, the YOLOv4 detector was also compared with its predecessor, YOLOv3, showing advantages in both speed and accuracy. YOLOv4 achieved higher accuracy while having faster processing speeds so it is ideal for real-time applications. This literature showcases how the combination of YOLOv4 and DeepSORT algorithms significantly benefits the ability to detect, track, and count multiple objects in real-time. For that reason, we can take their methodology into consideration to perform the same tasks on vehicles in our project.

The next paper titled "Vehicle Counting at Road Intersections Using Video Data and Zone-Based Method with Additional Contour" by Ibryaeva et al. (2020) addresses the assignment of counting vehicles and determining their movement directions at road intersections [2]. The authors propose an innovative method that incorporates an additional inner contour to improve the accuracy of vehicle counting by expanding the zones of interest, thus increasing the likelihood of detecting fast-moving vehicles. Their methodology combines the YOLO (You Only Look Once) object detection network with the CSRT (Channel and Spatial Reliability Tracker) from the OpenCV library for keeping track of vehicles at intersections. YOLO excels at quick object detection by analysing an entire frame simultaneously, while the CSRT tracker maintains vehicle trajectories over time. Nonetheless, the novelty of the project lies in the addition of an inner contour within the zones of interest. This improvement supports detecting vehicles traveling at high speed through the outer contour which affects the accuracy of the counting process. The operation starts with constructing an inner zone based on the geometrical features of the outer area. This is done by calculating the intersection points of diagonals and utilizing a scaling factor to define the inner region. The inner contour is responsible for ensuring that vehicles that may bypass the outer contour due to speed or delayed detection can still be tracked as well as counted accurately. Regarding the implementation phase, the researchers apply their structure to analysing real-world video data collected from a local intersection, including scenarios with varying conditions such as daytime, nighttime, plus rain. After that, they evaluate the accuracy of two algorithms: one using only the outer contour and one using both the outer and inner contours. As a result, the paper demonstrates that the addition of the inner contour significantly reduces the Absolute Percentage Error (APE) in vehicle counting, particularly in challenging conditions. For instance, in the rain scenario, the APE dropped from 39.64% to 27.97% with the inclusion of the inner contour. The authors acknowledge that although the error rates are still relatively high due to limitations in the YOLO network and CSRT tracker, the inner contour substantially enhances performance in comparison to methods without it. This article allows our team to visualize the structure of the vehicle tracking and counting system which is a main component of our project. In addition to that, the findings also present a technique to yield better results.

In the third article 'A Real-Time Vehicle Counting, Speed Estimation, and Classification System Based on Virtual Detection Zone and YOLO' by Lin et al., the researchers document the development of an intelligent traffic monitoring system aimed at enhancing vehicle detection, and classification, along with speed estimation in real-time [3]. The system incorporates several advanced computer vision technologies, including the You Only Look Once (YOLO) object detection model, a virtual detection zone, accompanied by Gaussian mixture models (GMM) to overcome challenges in traffic monitoring such as occlusions, environmental changes, and the need for real-time processing. The authors highlight that computer vision-based approaches, particularly deep learning algorithms, have demonstrated promising results in vehicle detection. Nevertheless, real-time traffic monitoring requires systems that can not only detect vehicles precisely but also classify them into different categories and estimate their speeds efficiently. The primary goal of the proposed system is to enhance vehicle counting and classification efficiency through the integration of YOLO, GMM, and virtual detection zones. According to Lin et al., their traffic monitoring procedure involves 3 main assignments: vehicle classification, counting, and speed estimation. In terms of counting with speed estimation, a GMM was used for background subtraction to identify moving objects in each frame, while virtual detection zones were employed to perform vehicle counting. The virtual detection zone captures a vehicle when it enters and exits the predefined zone, allowing for an accurate count. Whereas the vehicle speed is estimated by calculating the distance traveled by a vehicle within the detection zone and dividing it by the time taken. For the next task, YOLOv3 and YOLOv4 models were used for classifying the detected vehicles into one of the six predefined classes. The authors select YOLO due to its proven capability of detecting multiple objects in real-time with high accuracy. YOLOv4, in particular, offers enhanced speed and accuracy compared to YOLOv3, making it well-suited for traffic monitoring applications. The proposed system was evaluated using three datasets: the Montevideo Audio and Video Dataset (MAVD), the GARM Road-Traffic Monitoring Dataset (GRAM-RTM), and a custom dataset collected by the authors. YOLOv4 achieved the highest classification accuracy, with 98.91% on the MAVD dataset and 99.5% on the GRAM-RTM dataset. Moreover, the system demonstrated robustness in different environmental conditions, achieving

99.1%, 98.6%, and 98% classification accuracy during daytime, nighttime, and rainy conditions, respectively. In addition to classification, the vehicle counting accuracy was also high, particularly during daytime (95.5%) and nighttime (98.5%). However, performance dropped slightly in rainy conditions (94%). The system's speed estimation capability was evaluated by comparing actual vehicle speeds with estimated speeds, and the results showed an average absolute percentage error of 7.6%.

On grounds of the examples as well as findings that have been synthesized from the literature above, our project team can decide the development for our vehicle counting module which is fundamental to traffic violation analysis. The YOLO models have showcased their excellence in object detection while zone-based counting strategies are applicable to our case.

Vehicle direction tracking

In the article 'Comparative Evaluation of SORT, DeepSORT, and ByteTrack for Multiple Object Tracking in Highway Videos', Abouelyazid provides a performance comparison of three state-of-the-art multiple object tracking (MOT) algorithms: SORT, DeepSORT, and ByteTrack [4]. The study aims to identify the most efficient algorithm for tracking vehicles and pedestrians in highway time-lapse videos, with implications for traffic monitoring, autonomous driving, and video surveillance applications. The three algorithms being analyzed are widely used in MOT: SORT (Simple Online and Real-Time Tracking), DeepSORT, and ByteTrack. SORT is a straightforward algorithm that uses motion cues and the Kalman filter to assign bounding boxes and track objects in real-time. DeepSORT extends SORT by incorporating deep learning techniques to reduce identity switches hence increasing tracking accuracy. Finally, ByteTrack, a one-shot detection-based approach, combines object detection together with tracking into a single model for enhanced efficiency. ByteTrack, SORT, and DeepSORT were tested using the same dataset and evaluation metrics to ensure a fair comparison of their performance. The metrics that the author used included Multiple Object Tracking Accuracy (MOTA), Multiple Object Tracking Precision (MOTP), Mostly Tracked (MT), Mostly Lost (ML), False Positives (FP), False Negatives (FN), and Processing Speed. Regarding the dataset, the study used an open-access highway video dataset as the input for three algorithms. The experimental results reveal that ByteTrack consistently outperforms both SORT and DeepSORT across most evaluation metrics. ByteTrack achieved a MOTA of 77.3% and a MOTP of 82.6%, significantly surpassing the results of DeepSORT (61.4% MOTA) and SORT (54.7% MOTA). ByteTrack also exhibited the lowest number of ID switches (558), indicating its superior ability to maintain object identities. When it comes to tracking completeness, ByteTrack tracked 54.7% of objects throughout their lifespan (MT) while having the lowest percentage of mostly lost objects (ML) at 14.9%. ByteTrack also demonstrated better detection accuracy with fewer false positives (3,828) and false negatives (14,661) than DeepSORT and SORT. Additionally, ByteTrack maintained the highest processing speed at 171 FPS, making it well-suited for real-time applications. In contrast, DeepSORT's processing speed was relatively low (61 FPS), potentially hindering its efficacy in time-critical scenarios.

The application of YOLO and ByteTrack algorithms is also documented in the paper 'Traffic Impact Assessment System using YOLOv5 and ByteTrack' by Ng et al. [5] The purpose of the system is to solve traffic congestion using advanced computer vision techniques: vehicle detection, classification, counting, and idling time analysis based on video input. The product proposed by the authors comprises a user interface for drawing lines and regions on video frames to define areas of interest, followed by automatic vehicle detection using YOLOv5. The system is capable of performing several functionalities: vehicle classification which is identifying different types of vehicles (cars, trucks, motorcycles, etc.) in real-time, vehicle counting as they pass through user-defined lines or regions as well as idling time measurement for calculating the idling time for vehicles in specific regions to assess congestion levels. The system was evaluated using two videos with a resolution of 1280x720 and a duration of two minutes each. Moreover, the study focused on comparing the tracking performance of ByteTrack, OC-SORT, and StrongSORT, using metrics like processing time and tracking accuracy. The tracking accuracy for the three algorithms showed similar results, with ByteTrack and OC-SORT demonstrating the fastest processing times. ByteTrack processed Video 1 in 3.37 minutes and Video 2 in 2.30 minutes, making it suitable for real-time applications. StrongSORT, on the other hand, was significantly slower, taking over 12 minutes for Video 1. In terms of vehicle counting accuracy, ByteTrack achieved a high accuracy score for line counting, reaching up to 100% in some cases. However, region-based counting exhibited some inaccuracies, mainly due to extra counting instances caused by incorrect detection of historical points.

It can be derived from these studies that ByteTrack and DeepSort are the most favorable among candidates for tracking transportation. This means we can narrow down the options to implement in our design, thus saving us time to complete other tasks throughout the project.

Plate recognition

The last research in this literature review is ‘Automatic Number Plate Character Recognition using Paddle-OCR’ by Sarkar et al. presents an Automated Number Plate Recognition (ANPR) system using the Paddle-OCR model [6]. The authors develop an ANPR system that implements Paddle-OCR, an open-source optical character recognition model. Paddle-OCR is known for its high accuracy and support for various languages and orientations. Regarding methodology, the application aims to automatically detect number plates, localize characters, and identify them accurately using image processing and deep learning techniques. The first task in the system’s flow is license plate localization in which a YOLOv8 model is responsible for detecting the number plate positions within the input images. Subsequently, several preprocessing techniques were applied, such as Gaussian adaptive thresholding, Otsu’s thresholding, morphological transformations, contour edge detection, and Canny edge detection so as to improve the clarity along with the contrast of license plate characters. In addition to that, the dataset was augmented using techniques like random rotation, zooming, shearing, and horizontal flipping. This step expanded the dataset to 1,106 images, improving the robustness of the model. Fine-tuning is also applied as the pre-trained Paddle-OCR model was fine-tuned using the custom dataset. The authors modified the model architecture by incorporating ResNet50 as the base model and adding linear layers for better character recognition. The training process employed the Cross-Entropy Loss function and the Adam optimizer for parameter updates. The fine-tuned Paddle-OCR model was tested against other OCR models like EasyOCR and TesseractOCR using five sample images from the augmented dataset. The Paddle-OCR model demonstrated superior performance, achieving an average accuracy of nearly 98-99%. In contrast, EasyOCR and TesseractOCR exhibited lower accuracy, especially with images of poor quality, small dimensions, or suboptimal lighting. The consistency of the Paddle-OCR model's results and its ability to handle blurred images underscore its effectiveness for ANPR applications. The study offers an alternative character recognition model to our current one which is the EasyOCR. A comparative evaluation is necessary so that our team can decide which model is appropriate for our system.

4 Solution Design

4.1) Original Scope of Work

For the scope of work, our project will consist of 2 parts, the video analyser backend and the website frontend. Both parts will be combined into our complete project.

The video analyser will have these core functions and features:

- The video analyser will have the ability to receive a video file of an illegal parking area to identify if there are any vehicles parking in that area longer than 5 minutes and return the vehicle's license plate and a picture of the vehicles and timestamp.

- The video analyser will have the ability to receive a video file of a one-way street or multiple-lane street to detect vehicles that deviate from their lanes or travel in the wrong direction on one-way streets and return a picture of the vehicles and timestamp along with license plate if possible.

- The video analyser will have the ability to receive a video file of a red light or stop sign area to identify vehicles that run over red lights or fail to stop at stop signs by analysing traffic camera footage and returning a picture of the vehicles and timestamp along with license plate if possible.

- The video will be analyzed using machine learning (ML) models and the result of the analysis will be returned within a sufficient time frame, while the time frame is not a priority, the team aims for not longer than 8 times the video length.

The website will have these core functions and features:

- A form so users can upload videos as well as mark which type of video (parking space, one-way street) they are uploading as well as the analysis function they want to use

- A login and register feature so user can make their own account and have their own database of uploaded videos. The user can see the analysed result of their previous video through the database.

- Intuitive and elegant UI that is easy to look and understand

- Informative Dashboard that contains the license plate of the identified vehicles as well as other metrics from the analysed video like vehicles counted, and classification of each vehicle (car, motorbike)

Originally, for evaluation, the project will have to satisfy these criteria:

- At least 2 out of 3 functions (illegal parking and red-light detection) will be complete and usable.

- The website must be fast to access with minimal delay (transition between pages is 3 seconds max except for from scan page to result in page which will be a max of 8 times the video length) along with all basic features completed and is able to be used for analysing video.

- Users will be able to fill in the form and input a video to receive the analysis result in a sufficient time frame.

- Time takes at most 8 times the length of the input video to complete the analysis. For this, we will log the time it takes to complete the analysis on the log

- Capture 85-90% of all violation vehicles and output all their pictures with timestamps of when they appear. To test this mile, we will use videos with an identifiable number of objects and compare analysis results with our results

- For vehicles whose license plate can be captured and analysed, license plate result is returned with 75% accuracy (at least 6 out of 8 characters are correct). Compare results from the analysis

- The time to switch between front-end pages is 2 seconds at the latest on most computers, we will measure this using the Python library to record the load time during the testing process

- The website can support the process of signing up, logging in, analysing videos, and saving previously analysed videos as history without issue. For this we will perform the following process 3 times using different computers: sign up, log in, analyse a 30-second video, and open that video in the history dashboard. This is considered a successful test when the process is complete within 5 minutes on all 3 computers.

- Website can continuously analyse 5 videos in a row without issue. For this, we will submit a 30-second video and wait until the analysis is finished, after which we will submit another video and repeat the process until 5 videos are submitted and analysed without issue. The video will be of different lengths and use different violation options for analysis.

4.2) Changes to scope:

- With advice from our company supervisor, we have shifted the focus of our project to first capture the picture and timestamp of vehicles that violate traffic law to support police officers with identifying violations in a long video. License plate recognition will now only be done on vehicles with clear and visible license plate

- Users will now have to draw their area of violation on a separate page after uploading their video, this is done since the area to detect violation of each video is different. It is advised that letting the user draw the area will help the area be the most accurate for each use case

- An area of detection was added to the website flow, this is to limit the number of vehicles tracked to those inside the detection area.

- Due to time constraints, we had to remove the login and register function, as well as the history function. This is to make sure that our main functions are completed on time.

- The changing of the project scope has led to the changing of the testing plan, with tests being modified or removed and new tests added in to fit with our new direction, the new test changes are discussed more in section 5.

4.3) Approach:

4.3.1) Analyser: For our analyser, we will split the process into 4 parts: traffic light detection, vehicle detection, license plate detection, and license plate text recognition seen in Appendix Figure 1. When a video is input, we will draw 2 regions of interest (ROI) on the video frame as seen in Figure 1 below. The first area is called the detection area, which is the big area, vehicles that enter the detection area will be tracked and labeled, this is to prevent tracking of vehicles from outside the necessary area. The second area is called the violation area, which is the smaller area, vehicles that enter the violation area will be marked as violating the traffic rules, and they will have their pictures taken along with timestamps and license plates if possible. When a video is input into the analyser, we will then use the ML model to detect vehicles inside the detection area. When we detect vehicles that violate the rules through the violation area, we will save and extract the image of that vehicle. We will then run the model for license plate recognition on the image to extract the area containing the plate of the vehicle. Finally, we use OCR to recognize the text of the plate and save it if a result is detected. The images and data will be saved on a database and displayed on the result page when the analysis is complete.

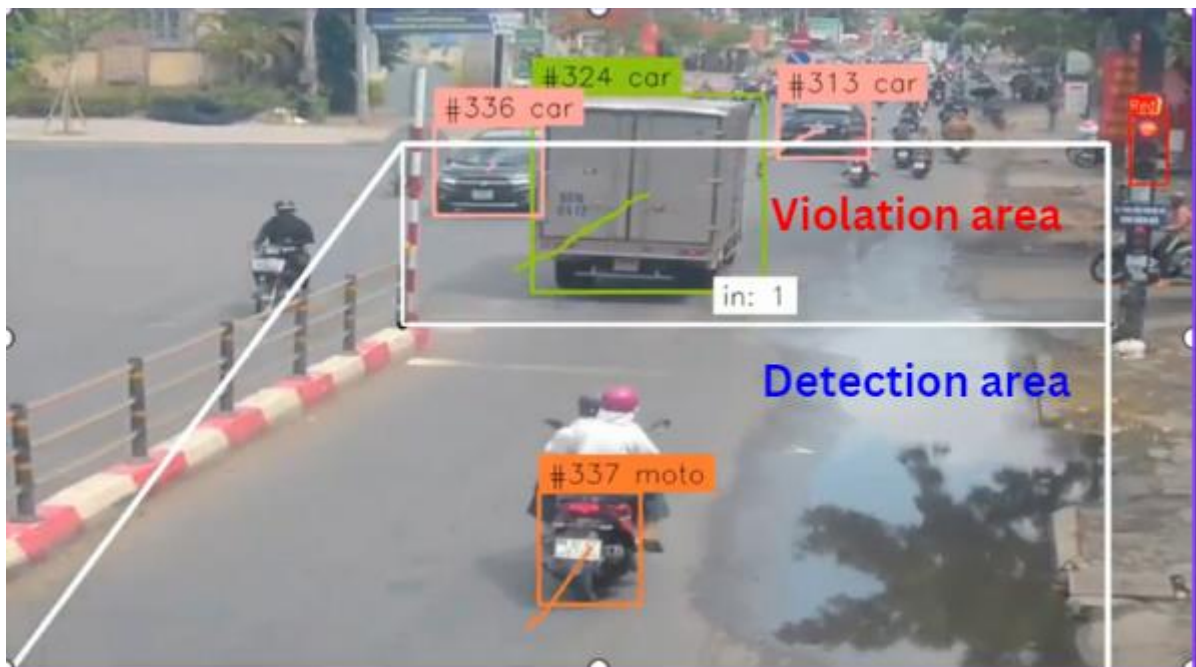


Figure 1: Picture of an analysed video frame

a) Vehicles detection and tracking:

Considered solution and chosen solution:

For the considered solution, we will compare between 3 models: detectron2, YOLOV8, and YOLOV9 for detection. When comparing we prioritize the available support and pre-trained models as well as accuracy. This is to make sure that our chosen model will be the most suitable for our task. The comparison is listed in the appendix (Model Comparison table).

The conclusion of our research as well as the testing process is that YOLOV8 is the leading model in terms of support and has a lot of pre-trained models available, it also has very fast training and inference speed without trading much of accuracy which makes it very suitable for our project. YOLOV9 while newer also has fewer pre-trained models and support and our testing has not resulted in notable improvement compared to V8. Detectron2 is more accurate and has more support and pre-trained models compared to V9, however, it is noted to be slower in training and inference speed compared to those of the YOLO family.

After real testing using all 3 models, we were able to see that for our project, YOLOV8 and YOLOV9 have easier setup and accuracy than Detection 2. When comparing YOLOV8 and YOLOV9 using trained models for vehicle detection, we have observed that YOLOV8 has a significantly shorter analysis time than YOLOV9 while retaining similar accuracy. Therefore, for our project, we use YOLOV8 for the vehicle detection model with the support of the ROBOFLOW supervision library. Our deeper comparison between YOLOV8 and YOLOV9 with data and conclusion can be seen in the YOLOV8 and YOLOV9 model comparison Table.

For tracking, we considered 3 solutions, Byte track, Deep sort, and Bot sort. After researching and doing our own tests. We have decided to use Byte track as our tracking algorithm, this is due to Byte track already integrated into YOLO making the accuracy and speed better than its competitors. With the support from the ROBOFLOW supervision library, we can easily tune the parameters to track vehicles better.

Solution Design:

The model will be used first to detect and label vehicles into 2 classes: motorbikes and cars, here we consider all non-motorbikes vehicles as cars for easy labeling. When vehicles enter the detection area, it will be marked as seen in Figure 2 below. The label includes the ID number assigned to the vehicle, and the class it belongs to along with the trace line, this trace line shows the path the vehicles have traveled and helps with vehicle tracking. As the vehicles are being tracked by the models, the class and id number of vehicles will not change as they move.



Figure 2: Label on a motorbike

As seen in Appendix Figure 1, when a video is uploaded to the server, depending on the chosen type of violation, the way the video is analysed will differ:

+Red light violation: For this detection mode, for every frame in the video. First, the vehicle detection model will be used to detect and track all vehicles in the detection area. All vehicles will be tracked until they leave the detection area. If the traffic light detection model returns a red light, then all vehicles that cross the violation area will be captured and have their picture taken along with their timestamp and license plate if possible. If the traffic light detection model returns a green light, then the vehicles will continue to be tracked without having their data recorded until they exit the detection area. This process will repeat with every frame until the end of the video, after which the collected data will be displayed on the result page in the form of a result dashboard seen in Appendix Figure 7.

+Wrong parking violation: For this detection mode, for every frame in the video. First, the vehicle detection model will be used to detect and track all vehicles in the detection area. When a vehicle enters the violation area, it will count the time the vehicle stays in the violation area. When the timer reaches 5 minutes, the vehicle parking in the violation area will be

captured and have their picture taken along with their timestamp and license plate. If the vehicle leaves the violation area before the timer reaches 5 minutes, the timer will be reset. This process will repeat with every frame until the end of the video, after which the collected data will be displayed on the result page in the form of a result dashboard.

b) Traffic light detection:

Considered solution and chosen solution:

For this model, we will compare between YOLOV8 and YOLOV9. This is due to the availability of pre-trained models and data for traffic light detection. After training pre-trained models with our own labeled traffic light datasets and comparing the accuracy and inference time, we decided that YOLOV8 would be our model of choice, this is because while both models perform similarly in terms of accuracy, YOLOV8 has much faster speed compared to the V9 counterparts and is also available in ROBOFLOW supervision for many support functions.

Solution design:

This model will be used for red light violation detection in our project. When a video is inserted, this model will continuously scan the frame for the nearest traffic light and return whether it is a red light or green light, if the result is red, we will capture all vehicles that cross the area marked by the violation area. This process is seen more clearly in Appendix Figure 1. The models will repeat this process on all frames of the video to make sure the light signal is always updated. We decide to skip the yellow light signal to lessen the runtime of our model. Figure 3 below shows how the model works in action, both for detecting red light and green light.



Figure 3: Red light detected (left) and green light detected (right)

c) Plate detection:

Considered solution and chosen solution:

For this model, we will compare between YOLOV8 and YOLOV9. This is due to the availability of pre-trained models and data for Plate detection. After training pre-trained models with our own labeled plate datasets and comparing the accuracy and inference time when we extract the license plate from test vehicle pictures, we decided that YOLOV8 would be our model of choice, this is because while both models perform similarly in terms of accuracy, YOLOV8 have much faster speed compared to the V9 counterparts.

Solution design:

For Plate detection, we use YOLOV8 to extract the license plate image from the vehicle image. From the test results, we have determined that YOLOV8 provides the most accurate result with the fastest running time of the 3 ML models. The

result of running this model on a vehicle image is seen below in Figure 4. This extracted image will be used for license plate recognition.



Figure 4: Extracted license plate examples

d) License plate text recognition:

Considered solution and chosen solution:

For This section, we consider 3 libraries along with 2 alternative solutions, the considered solution are as follows:

- + EasyOCR: EasyOCR is a Python-based PyTorch library that falls upon good GPU to show accurate results. There are three main components feature extraction, sequence labeling, and decoding. It doesn't have many software dependencies like Tesseract OCR, it can directly be used with its API. The main feature of easy OCR is that it reads characters from images and returns the coordinates where it was located.

- + Tesseract OCR: Tesseract is an open-source optical character recognition Engine. It can be used directly or by using an API to extract letters and numbers from images. Compared to easy OCR, it supports fewer languages. However, Tesseract OCR has been noted to be best in detecting the letters from the writing as compared to EasyOCR.

- +Paddle OCR: Paddle OCR is an open-source OCR (optical character recognition) toolkit developed by PaddlePaddle, an AI and deep learning platform. The Paddle OCR toolkit is designed to be a fast and accurate OCR solution for recognizing and extracting text from images and documents.

Alternative solution:

- + Train CNN model for text recognition: for this approach, we will train a CNN model on a Vietnamese license plate letters and number dataset, this model when run on a frame will detect all the letters and numbers on it, and we will then remove all noise and incorrect detection and connect the rest into the license plate and output it to the dashboard.

- +Commercial online API: like plate recognizer to handle the plate detection and text recognition, in this scenario, our team project will focus on extracting violation images from a video using a vehicles detection ML model to maximize unique objects detected and do quality improvement to those images. The images will be sent to a commercial online API and the result will be processed and displayed on the dashboard. While this solution guarantees high-accuracy license plate text recognition as long as our picture quality is high as well as reducing the models that we need to train thus reducing training time, it will cost more money as we have to subscribe to their service.

After considering all available solutions we have decided on PaddleOCR for its high accuracy and speed. Tesseract OCR when tested on our data is lacking in accuracy and speed, and the library does not normally separate 2 lines license plates. EasyOCR is similar in speed to our top option and can separate 2 lines of license plates but has lower accuracy only around 60% compared to PaddleOCR 80%. The alternative solution was considered but due to the lack of training time and budget, we were not able to do any tests regarding them. The comparison table is shown below.

OCR model Comparison table

Solution	Advantage	Disadvantage
PaddleOCR	+Fastest speed in non-alternative solutions +Good accuracy (around 80%) +Can read 2-line license plate +Can read many blurry pictures	-Need training time -Need more set-up time -No detailed guide available
EasyOCR	+Decent accuracy (around 60%) +Can read 2-line license plate +No training time needed	-Slow -Have trouble with 2-line license plate -Have issues with blurry picture
Tesseract OCR	+Decent accuracy (around 60%) +No training time needed	-Cannot read 2-line license plate without setting up -Slowest -Have issues with blurry picture
Train CNN model	No test was done due to lack of time.	
Commercial online API	+Best accuracy +Very fast +Can read 2-line license plate +Can read many blurry pictures	-Cost money -Limited usage -Require set-up time and integration time

Solution design:

For our solution, first, we employ a pre-trained YOLOV8 model for the purpose of identifying license plates inside a picture. The fast capability and high accuracy of YOLOV8 guarantee rapid identification, rendering it well-suited for our needs. The bounding box coordinates of the identified license plate are used to crop the license plate area from the input picture with high precision.

After the license plate image is cropped, we use image preprocessing so that iterative improvements are made to the cropped license plate to boost OCR accuracy. The procedures include scaling, implementation of Gaussian blurring to minimize noise, and conversion of the picture to grayscale. This helps enhance the image clarity, especially for lower-resolution photos.

When the photo finishes processing, we use the PaddleOCR platform, when setting the parameter `use_angle_cls=True`, which is used for extracting text from both the upper and lower sections of the license plate. This OCR model can handle tilted or slanted text, which is typical in real-world license plates. PaddleOCR identifies and extracts the text, which is subsequently cleaned using regular expressions to eliminate undesirable characters.

After PaddleOCR finishes its analysis, we put the result through formatting and character conversion. This is because the recovered text commonly includes OCR misrecognitions, such as mistaking 'O' with '0' or 'A' with '4'. To address this, we apply character mappings, transforming these misrecognized characters back to their proper form using the standard format of Vietnamese vehicle plates. The top and bottom lines are formatted individually using precise standards, ensuring the license plate maintains the intended layout.

After the conversion stage, we put the newly converted text through text validation. In this phase, we verify the retrieved text to see if it complies with conventional license plate formats. The top line is verified based on particular standards, such as having two numbers followed by a letter, while the bottom line must have 4 or 5 digits. This validation stage is critical for filtering out inaccurate or partial readings from the OCR process. The final result of this module is seen in the figure 5 below.

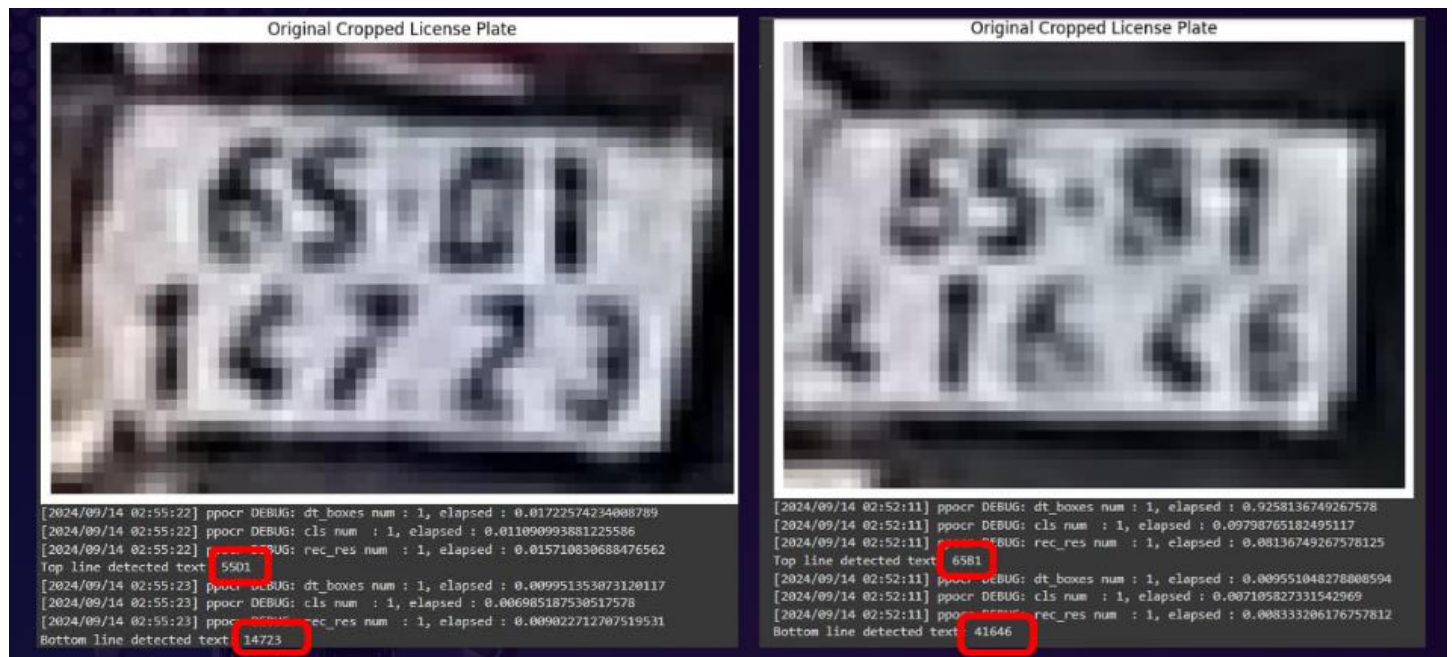


Figure 5: OCR results on a license plate image

The OCR module is scalable for bigger applications and may be modified to accommodate license plates from multiple nations by altering the OCR settings and validation procedures. With its modular construction, components like the detection model or OCR engine may be simply switched or modified, allowing flexibility for future upgrades or unique use cases.

c) Website design:

For our website, in our original approach, users accessing the website will see the homepage. From the homepage, they will navigate to the account page so they can log in or sign up, after that, they will be moved to the upload page to upload a new video. When choosing to upload a new video, the user will be presented with a form so the user can upload the video, name it as well, and mark which type of violation (parking space, red light) they are uploading, and the analysis function they want to use. When choosing to view past results, users can view the analysed video as well as the dashboard containing previous results. Our website prototype workflow can be seen in the appendix Figure 2.

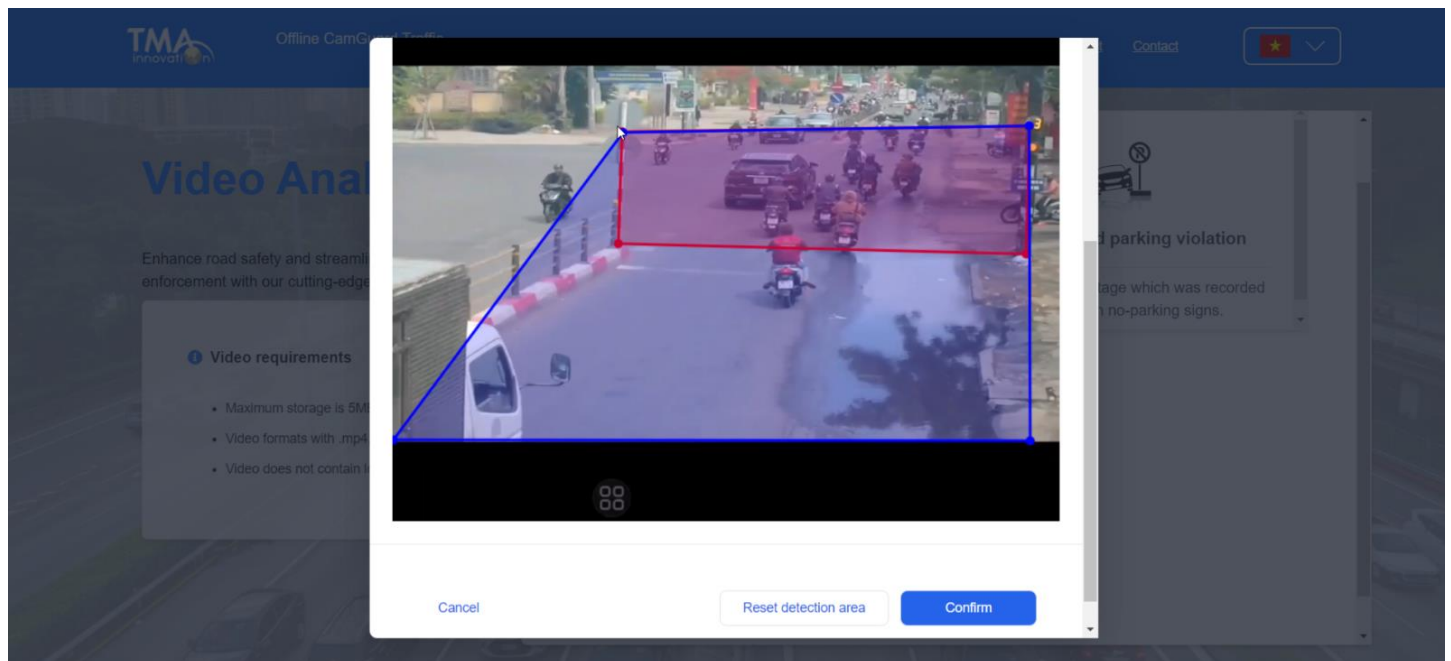


Figure 6: Detection drawing page

Due to time constraints, we decided to remove the user login and register functions as well as the history function. With advice from our supervisor, we have added the area drawing pop-up after uploading the video so users can draw their own detection and violation areas. This is due to each video may have been filmed from a different location and angle so the area in each video should be left for the user to decide, the new page can be observed below. In Figure 6 above you can see that the user draws their own detection area (red) and violation area (blue). Each area can have a maximum of 10 points, this is so users can shape the areas to the area that they need. After the user designated the areas, the coordinate will be saved and sent to the backend analyser. The user can select an area consisting of less than 10 points by clicking near the first point, doing so will connect all drawn points to make a custom area with less than 10 points.

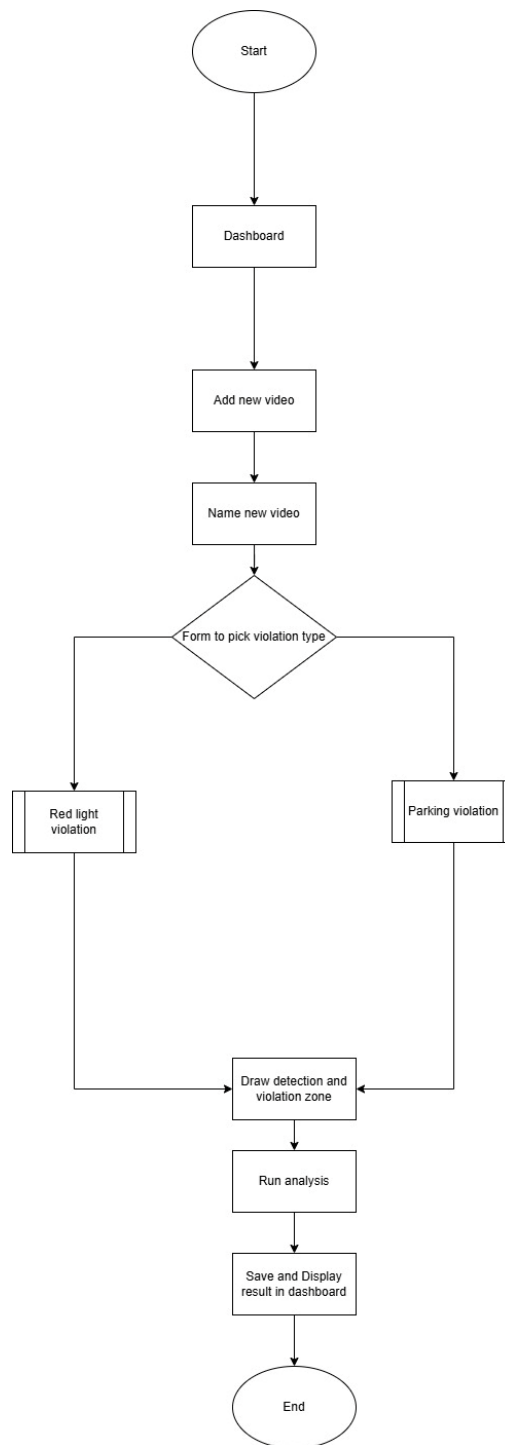


Figure 7: Website flowchart

We've chosen ReactJS to develop our final website. ReactJS is a highly efficient JavaScript library for building user interfaces, particularly due to its component-based architecture, which allows developers to create reusable, modular UI elements. This design promotes cleaner, more maintainable code. React's virtual DOM optimizes performance by minimizing direct interactions with the actual DOM, ensuring faster rendering and smoother user experiences. Additionally, its unidirectional data flow simplifies the management of the application state, improving both predictability and ease of debugging. With a vast ecosystem, strong community support, and seamless integration with tools like Redux and React Router, ReactJS is an excellent choice for developing scalable and maintainable applications in modern web development. After consideration and consulting with industry partners, our new website flow can be observed in Figure 7 above.

5 Result analyses & discussions

5.1 Completed deliverables for analyser:

- The video analyser has the ability to receive a video file of an illegal parking area to identify if there are any vehicles parking in that area longer than 5 minutes and return the vehicle's license plate, a picture of the vehicle, and a timestamp.
- The video analyser has the ability to receive a video file of a red light or stop sign area to identify vehicles that run over red lights or fail to stop at stop signs by analysing traffic camera footage and returning a picture of the vehicles and timestamp along with license plate if possible.
- The result of the analysis will be returned within a sufficient time frame; the recorded time is no longer than 8 times the video length when using 1 T4 GPU.
- 2 out of 3 functions (illegal parking and red-light detection) will be complete and usable.
- The final vehicle detection and tracking model will have an accuracy of over 75%.
- Licence plate recognition model achieves an accuracy of at least 75% (at least 6 out of 8 characters are recognized in most cases as seen in figure 5 and appendix figure 3)
- Image, data, and video are saved on a database without issue

5.2 Completed deliverables for the website:

- Intuitive and elegant UI and homepage that is easy to look and understand (Appendix Figure 4)
- A form so users can upload videos as well as mark which type of violation video (parking space, red light) they are uploading as well as the analysis function they want to use (appendix Figure 5, 6)
- The ability for users to draw their own detection and violation area, with 10 points limit for each area (website design Figure 6)
- Informative Dashboard that contains the license plate of the identified vehicles as well as other metrics from the analysed video like vehicles counted, and classification of each vehicle (car, motorbike) (Appendix Figure 7)
- Quick load time taken to switch between pages
- Switch time between pages is 2 seconds at most ensuring a smooth experience

5.3 Test taken and results:

a) Test in original scope

-Time takes at most 8 times the length of the input video to complete the analysis. For this, we have logged the time it takes to complete the analysis on the log using the time library and then compared the results obtained from it to the real video time. This test has been repeated multiple times with video of different lengths to ensure accuracy, this result is achieved when the analyser is run using a T4 GPU, the time taken gets longer when using a slower GPU or CPU with the slowest recorded time being 6 hours for a 1-minute video, while faster GPU gives faster results, with 2 T4 GPU returning time is at most 5 times the video length.

-Capture 85-90% of all violation vehicles and output all their pictures with timestamps of when they appear. To test this milestone, we will use videos with an identifiable number of objects and compare analysis count results with our results from the video.

-For vehicles whose license plate can be captured and analysed, license plate result is returned with 75% accuracy (at least 6 out of 8 characters are correct). For this test, we use the results returned from our OCR module and compare them to the original plate we have extracted.

-The time to switch between front-end pages is 2 seconds at the latest on most computers, we measure this using both the time Python library to record the load time during the testing process and real-time measurements. This test has been

completed with successful results, time to analyse video is excluded in this test, and have been repeated on multiple computers for consistent results

-The website can support the flow process without issue. For this we will perform the following process 3 times using different computers: Enter the home page, continue to the analysis page, select a violation type and upload a 30-second video, analyse the 30-second video, and open that video in the result dashboard. The process is complete within 5 minutes on all 3 computers.

-Website can continuously analyse 5 videos in a row without issue. The videos used are of different lengths and use different violation options for analysis.

b) New tests

- Draw a detection and violation area with an increasing number of points for each from 3 to the maximum amount of 10 points, if the coordinate of the area drawn is saved the test is considered complete. This test has been completed successfully without issues.

- Upload a video and check the Minio storage. If images of violation vehicles and the uploaded video are present, then the test is considered complete. This test has been completed successfully without issue.

5.4 Discussion on project progress:

-We were able to complete the analysis part of the project as well as the website flow

- The completion of the main scope of the project is done even with the added function of a user-defined detection area and violation area

-Most of the tests we were able to conduct were a success, aside from tests related to parts of the original scope we have not completed yet

- The model accuracy was within expected ranges, with results of around 80% of vehicles captured during normal time, and around 70 during times with busy traffic

-Due to the changing of scope and new understanding of the project requirements, new tests have been added to accommodate for those changes, the changes include:

+ Testing the user-defined drawing of the detection and violation area, this is to make sure that users can easily create the area that fits the video they want to analyse

+ Testing the Minio storage, this test is to make sure that our storage system is working properly as it is a very important component of our project

-Due to time constraints and issues while developing and training the AI models, many of our scopes have to be removed, these scopes include:

+Violation function: Reduced from 3 to 2 red light violations and parking violations, with the removed function being wrong lane violation, however as the minimum requirement only requires 2 out of 3 functions, this is within the expected range

+Register and Login function: Due to the time spent on the training of our models and the integration process, the user account portion of our project has been removed from the scope

+History function: Despite being able to save the resulting video on the server, since the user account portion was removed from scope, the history function was removed as well

+The removal of the user account and history function is an oversight of our planning process, this happened because we underestimated the time taken to train the models to a high accuracy, the number of models we have to train as well as the time needed for integration between website and analyser.

5.4 Potential and issues for future work:

- Our project has the potential to be expanded to many types of violation cases aside from the 2 cases we have completed, these cases include:

+Wrong lane violation: In this case, the analyser will detect if a vehicle is going in the reverse direction from the lane, the tracing on each vehicle will be used here to provide the direction the vehicle is moving, and the violation area will be used to capture the vehicle that moves in the wrong direction

+Multiple lane street: In our project, our analyser can only work on one lane at a time using the detection area; by using another layer of the ML model to identify different lanes, we can perform analysis on multiple lanes at once, allowing the analyser to work on streets with many lanes.

-Aside from potential violation cases, our project also has issues that can be expanded on in future work:

+Analysis speed: While the analysis speed has been measured in the test as around 8 times the length of a video. The test was conducted using a dedicated T4 GPU, the speed of analysis gets significantly slower on a slower GPU or CPU. During future work, there is potential to reduce the analysis time so that the project can run faster on slower GPU and CPU.

+Vehicle detection Accuracy: The accuracy of our analyser can be affected by heavy traffic and bad weather. This is an issue for us as we lack the training data for more specific weather and heavy traffic cases. In future work, it should be a priority to collect more training data from many different conditions of weather and traffic. It is also advised by our industry advisors to train different specialized models for more specific use cases and switch between them based on the condition of the inputted video.

+Vehicle tracking accuracy: When tracking vehicles, if the vehicles overlap with one another, the tracking will be bugged, and the vehicle will no longer be tracked. In future work, the tracking algorithm should be improved so that it can cover overlapping vehicles without losing track of them

+ Removed function: In future work, removed functions from our project such as the user account and history function can be added to improve this project. The account function and history function are both great additions to our project as they allow users to save data and review past videos when needed. The account system also provides extra security to the video data. User data can be stored and queried when needed using a scalable database like POSTGRESQL or MYSQL, while the history function could be implemented with the video already being saved in the Minio database

+ Online deployment and storage: Our project only works locally at its current state but there is potential for it to be deployed online using AWS or another cloud service. In that scenario, our Minio storage is programmed to be easily switched to AWS S3 storage to switch our storage online as well.

+ Improve OCR accuracy: Our OCR module accuracy can be improved in future work so it can more accurately recognize even bad-quality license plate images. This can be done by improving the model with more training data from different license plate

+Vehicles model detection: Currently our project can only sort detected vehicles into 2 classes cars for 4-wheeled vehicles and motorbikes. In the future, we would like to improve our analyser to be able to recognize and classify vehicles into different models. This could be done by training an extra ML model using labelled data of different vehicle models in Vietnam.

+ Image quality improvement: When extracting license plate images from vehicles, the image can sometimes be blurry and cause the OCR module to be unable to recognize the characters. In the future, we plan to research and implement better tools such as image enhancement libraries or AI to make the extracted image clearer

6 Summary & reflection

Teamwork and Communication:

- **What Worked Well:** Weekly team meetings were held consistently with all members attending, progress and issues were clearly conveyed to the stakeholders, and goals were clearly set and understood by both the team and the shareholders. Team communication is kept consistent through group chat to make sure team members are catch up to each other's progress. We have also made a note of every consultation with our shareholders and shared it on the team chat, all unavailable team members in each meeting are advised to check the notes at the end of the workday to catch up to the team's progress.
- **Areas for Improvement:** While weekly meetings with shareholders go well. Due to some team members' time constraints, not all members are up to date with the team's progress through the rest of the week, this causes our teamwork to be lacking. Due to our busy schedule, not all members can be present during the meeting with company supervisors, this also causes delays in our project as we have to catch those members up. In addition to that, we also use multiple channels for communication including MS Teams, Skype, and Facebook Messenger to meet our shareholders' preferences. As a consequence, there might be some instances where information is not consistent or missing.

Technical Approach:

- **What Worked Well:** Dividing the project into smaller, manageable tasks during the timeline visualization allows for parallel development and more efficient code reviews, both capstone A old timeline and capstone B new timeline can be seen in appendix. By choosing well-documented and industry-standard development tools for our project we have minimized technical roadblocks due to the comprehensive documentation and support of those tools.
- **Areas for Improvement:** Due to our unfamiliarity with ML and its development tools, we had trouble navigating and using ML-related tools at first, and that has slowed down our progress. Besides, most machine learning processes require powerful GPUs that are not readily available to us or not easily accessible. Therefore, this may hinder our progress regarding machine learning tasks. Moreover, the data sources of traffic surveillance recordings in Vietnam that can be obtained publicly do not satisfy desirable quality or other crucial requirements. As a result, the robustness of our AI models may be limited.

Project Management:

- **What Worked Well:** Clearly defined project goals and a well-documented timeline provided a clear foundation for development. Regular team meetings allowed us to identify and address potential issues early. The risk assessment we have made and improved have allowed us to identify and mitigate risks during Capstone B (Risk assessment table appendix)
- **Areas for Improvement:** The initial project timeline proved to be overly optimistic due to underestimating the complexity of certain features. Some of our processes are not carefully documented in terms of comparative evaluation. As our development entails a lot of trials and errors in order to finalize the best-suited structure, insufficient documentation of all methods we have attempted may lead to inadequate reflections of our effort.

These reflections provide valuable insights for the next phases of our project. By improving communication between team members, utilizing consistent communication platforms, prioritizing the time needed to learn a tool, and implementing a more iterative development cycle, we are confident in improving our efficiency and delivering a high-quality software solution that meets our project goals.

7 References

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8 Appendices

Link to initial project timeline: [Project Timeline.xlsx](#)

Link to new project timeline: [Project Timeline\(CAPSTONE B\).xlsx](#)

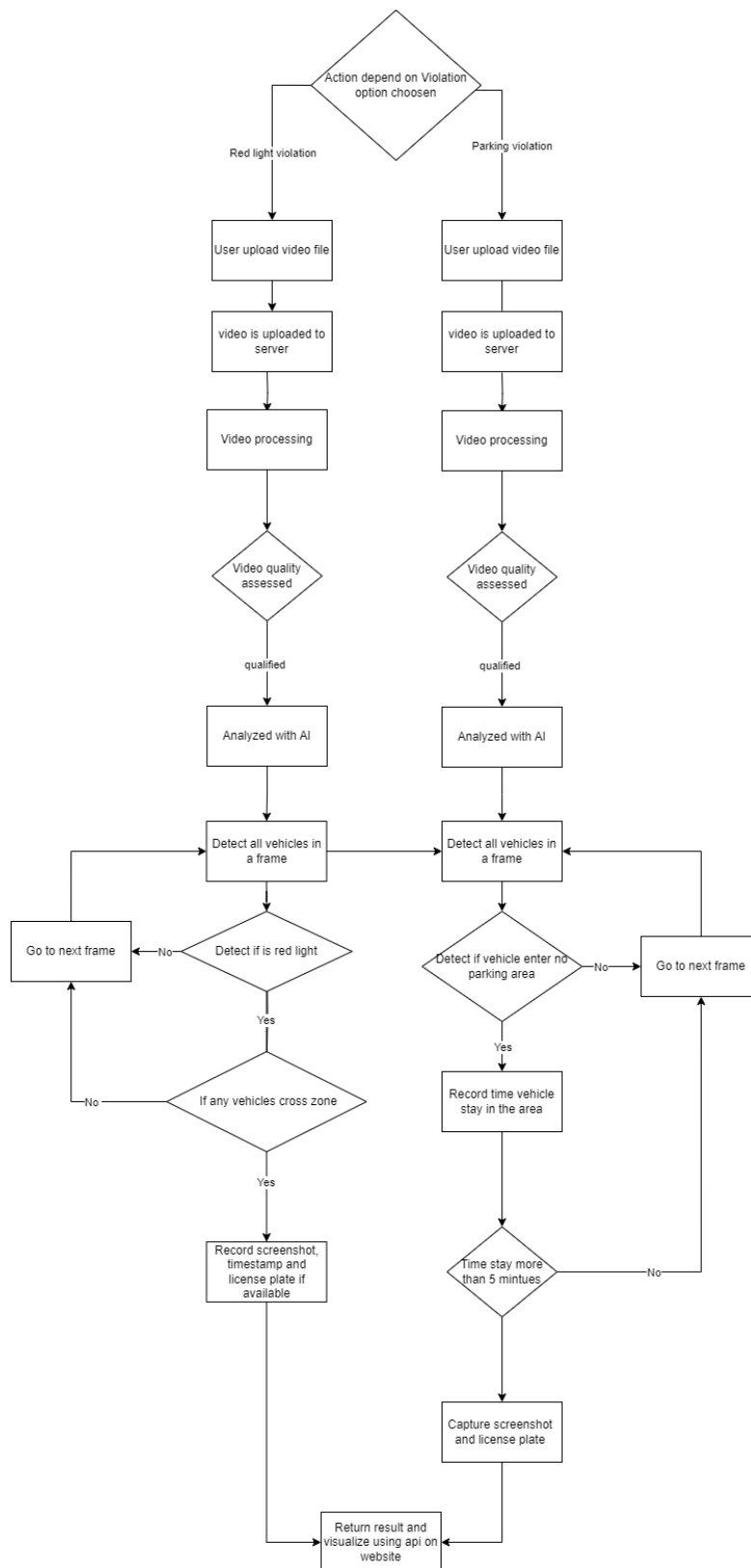


Figure 1: Video analysis workflow

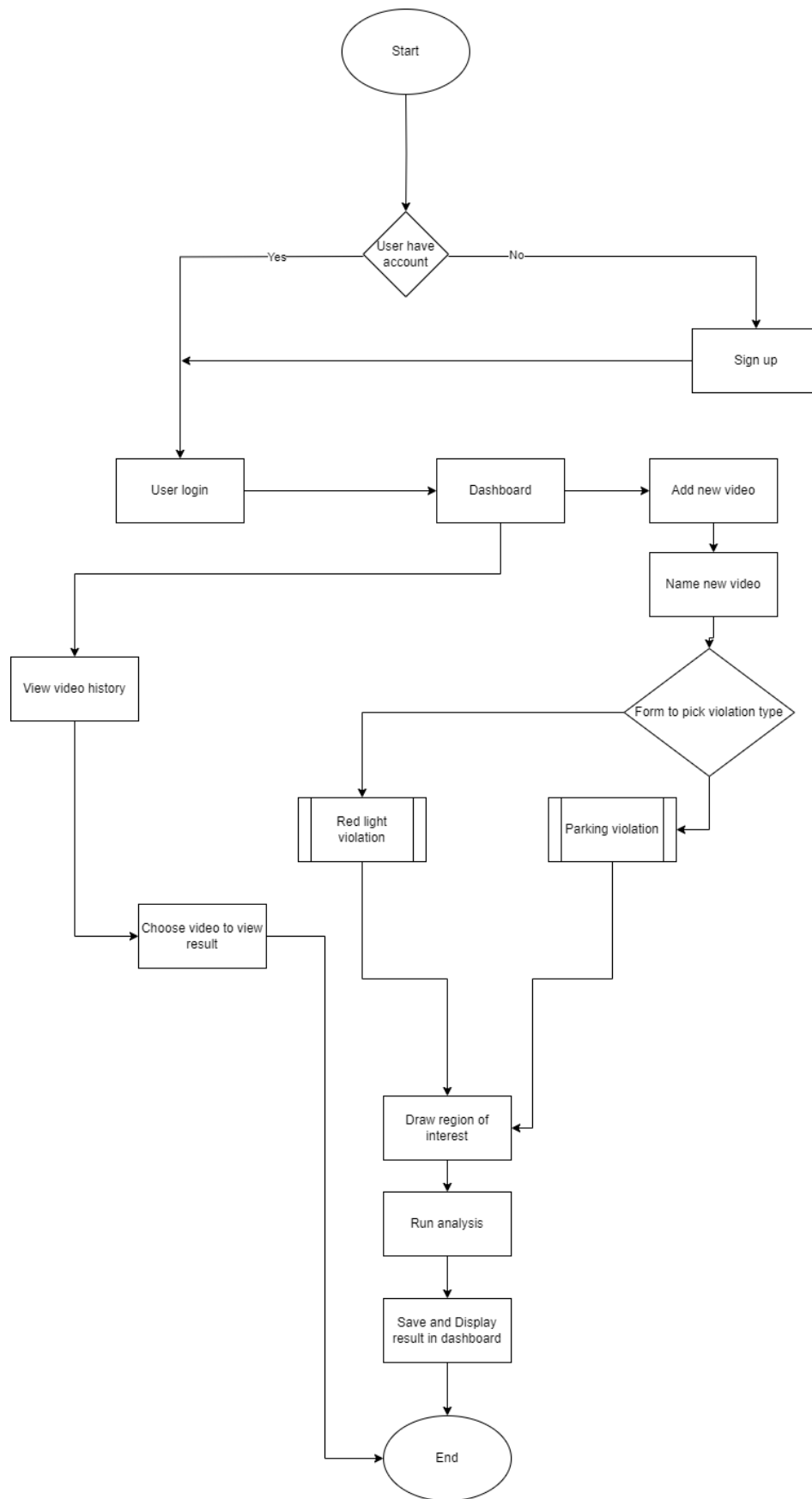


Figure 2: Previous project workflow

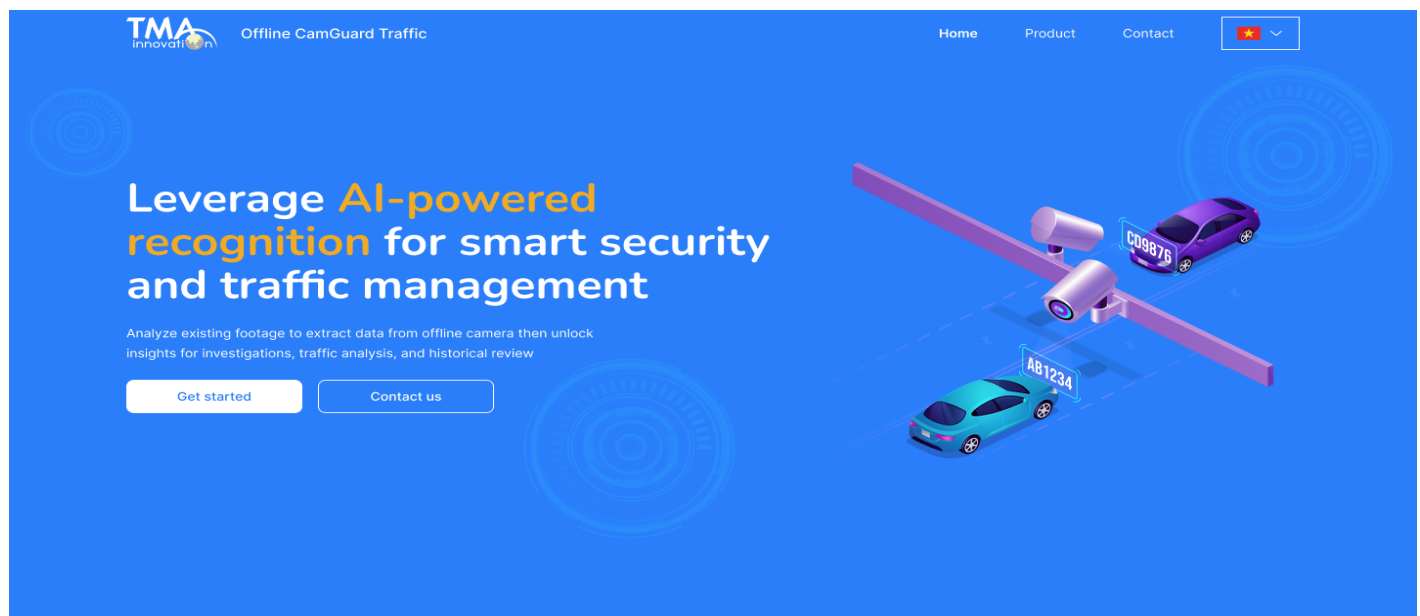


Figure 4: Homepage

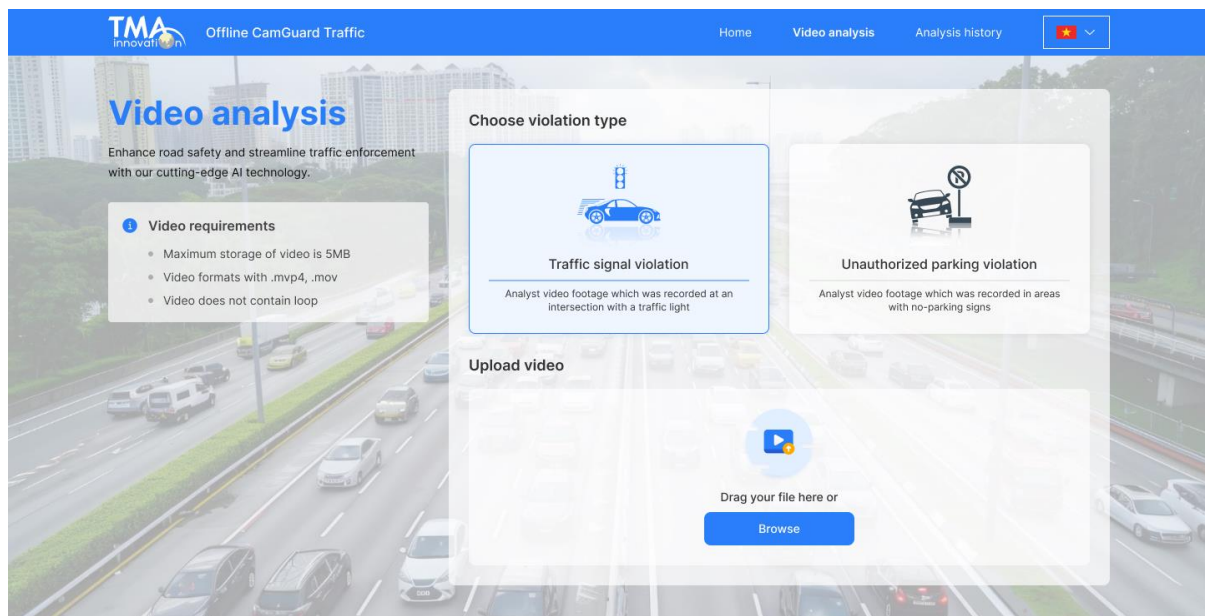


Figure 5: Process page option 1

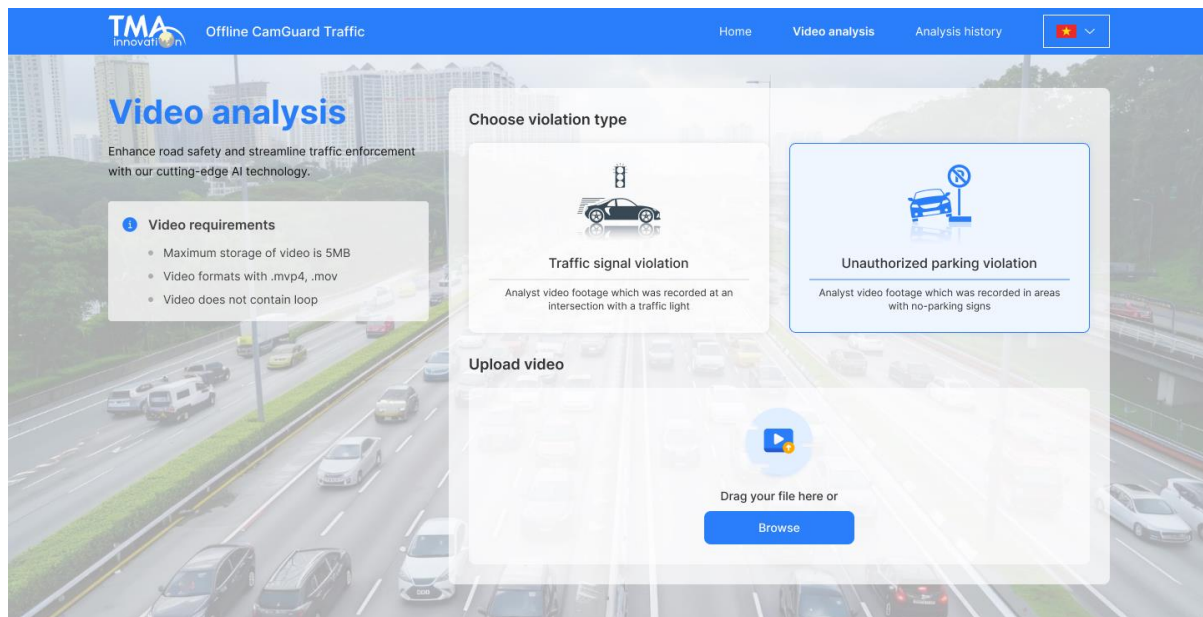


Figure 6: Process page option 2

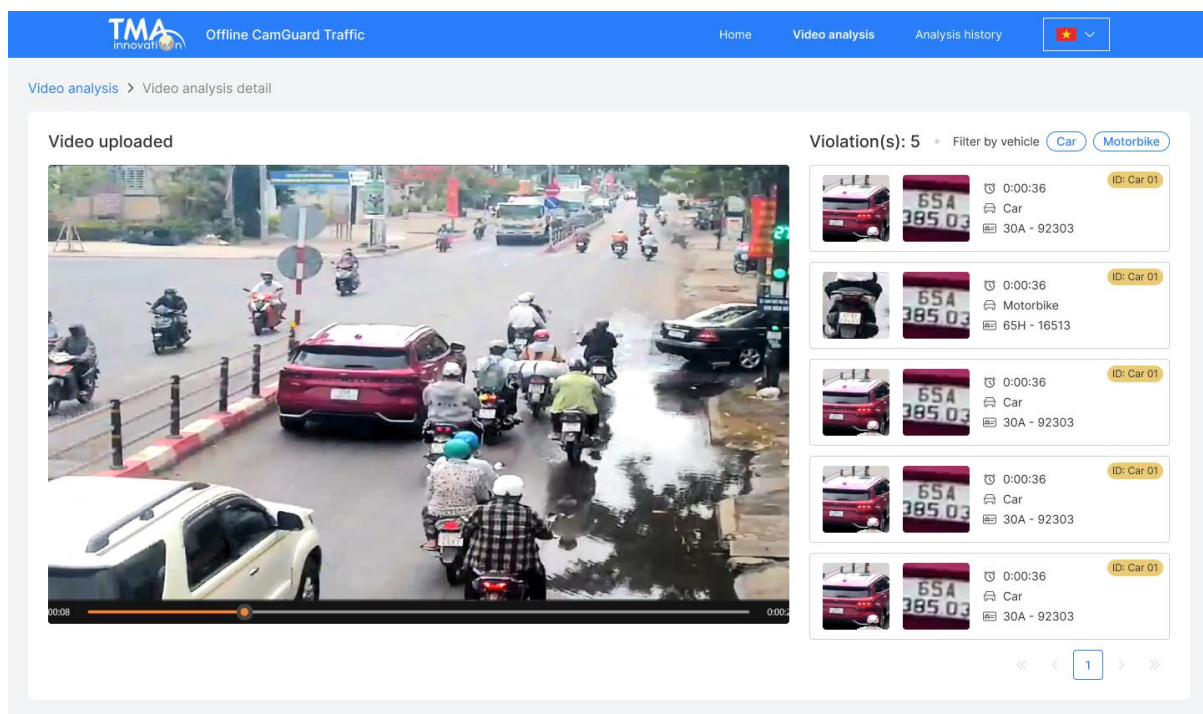


Figure 7: result page

		Consequence				
		Negligible 1	Minor 2	Moderate 3	Major 4	Catastrophic 5
Likelihood	5 Almost certain	Moderate 5	High 10	Extreme 15	Extreme 20	Extreme 25
	4 Likely	Moderate 4	High 8	High 12	Extreme 16	Extreme 20
	3 Possible	Low 3	Moderate 6	High 9	High 12	Extreme 15
	2 Unlikely	Low 2	Moderate 4	Moderate 6	High 8	High 10
	1 Rare	Low 1	Low 2	Low 3	Moderate 4	Moderate 5

Figure 8: Risk assessment grading

Model Comparison table

Models	Description	Comparison
Detectron 2	Detectron2 is Facebook AI Research's next generation software system that implements state-of-the-art object detection algorithms. It is a ground-up rewrite of the previous version, Detectron, and it originates from mask rcnn-benchmark	YOLOV8 = lots of pre-trained models +generally, more accurate given enough time -slower detection speed - less tutorials and online support YOLOV9 +more pre-trained models
YOLOV8	YOLO, short for You Only Look Once, made its debut in a 2015 research paper authored by Joseph	Detectron2: +Faster while not trading much of accuracy +More support

	Redmon, Santosh Divvala, Ross Girshick, and Ali Farhadi. Since its conception, YOLO has evolved and gone through several iterations, with each subsequent version building upon the advancements of its predecessors. YOLOV8 is the 2021 iteration of YOLO	+Is pre-trained for cars and motorbikes +Consume Fewer resources for running +Less training time -lower accuracy ceiling YOLOV9: +more pre-trained models +more support + higher true positive count - Higher false positive count - Tends to over-detection -Older technology
YOLOV9	The 2024 iteration of YOLO, with the introduction of 2 new technologies: Generalized ELAN (GELAN) architecture and Programmable Gradient Information (PGI)	YOLOV8: +Less false positive count - higher false negative count. -May miss objects Detection2: =speed +Consume Fewer resources for running +Less training time + Not much accuracy trade-off -Less pre-trained models and support due to new technology

YOLOV8 and YOLOV9 model comparison Table**Model and Training Details**

Parameter	YOLO v9	YOLO v8
Layers	467 layers	268 layers
Parameters	25,412,502 parameters	68,125,494 parameters
GFLOPS	102.5 GFLOPS	257.4 GFLOPS
Speed	1.50 it/s	1.85 it/s

Performance Metrics

Metric	YOLO v9 (All)	YOLO v9 (Green)	YOLO v9 (Red)	YOLO v8 (All)	YOLO v8 (Green)	YOLO v8 (Red)
Images	103	103	103	103	50	53
Instances	182	91	91	182	91	91
P (Precision)	0.935	0.917	0.952	0.932	0.904	0.960
R (Recall)	0.824	0.824	0.824	0.841	0.791	0.890
mAP@50	0.908	0.885	0.931	0.921	0.880	0.962
mAP@50-95	0.525	0.494	0.556	0.564	0.510	0.619

Observations :

Accuracy: YOLO v9 exhibits greater accuracy for both classes (Green and Red) compared to YOLO v8. However, the difference is modest (YOLO v9 has somewhat greater accuracy for the Green and Red classes).

Memory: YOLO v8 indicates somewhat greater memory for the Red class compared to YOLO v9. For the Green class, YOLO v9 performs marginally better in terms of recall.

mAP@50: YOLO v8 has a little better mean average accuracy at 50% IoU (0.921 vs. 0.908). YOLO v8 looks to perform better, notably on the Red class, where its mAP@50 is greater.

mAP@50-95: YOLO v8 also performs better over a variety of IoUs (0.564 vs. 0.525 for YOLO v9). This measure implies that YOLO v8 is better suited for recognizing items at varied degrees of overlap.

Speed: YOLO v8 is somewhat quicker in terms of inference speed, processing at 1.85 it/s compared to YOLO v9's 1.50 it/s. This discrepancy may stem from YOLO v9's larger parameter count.

Conclusion:

YOLO v8 looks to have higher overall performance, notably in terms of recall and mAP measures and is somewhat quicker in inference.

YOLO v9 demonstrates improved precision, but YOLO v8 exceeds it in recall and detection accuracy across numerous IoUs.

Risk assessment table

The Table grading is done based on Figure 8 table

ID	Risk name	Severity	Probability	Rating	Mitigation plan
1	Cannot complete milestones on time	4	2	8	-Weekly progress checking -Adjust the plan to account for the late delivery if needed
2	Cannot integrate the system in time	3	2	6	-Build system using modular design - Doing frequent integration testing during development
3	Model accuracy does not reach the expectation	3	2	6	-Find more data to train the model -use pre-trained models as a base for training
4	Member's computer unable to function	5	1	5	-Monitor computer status and perform maintenance regularly -Request company-supplied computer for backup
5	Lost of data due to corruption or error	4	1	4	-Regularly back up data -Save an offline version weekly

6	The deliverables do not meet stakeholders' requirements	4	2	8	-Weekly meeting with supervisors from the company -Regular progress updates and daily reports -Refer closely to the scope of the project
7	Exceed the budget	3	1	3	-Conduct thorough project estimation and budget planning -Monitor project expenses closely and track them against budget targets
8	Model Bias and Fairness	3	2	6	- Variance in training data and testing data. - Test a variety of models to pick out the best ones
9	User Experience and Usability Issues	3	2	6	- Act as a tester/user and thoroughly test the UI (the website)
10	Intellectual Property Issues	4	1	4	- Double-check the legal status of every resource that we use. -
11	Cybersecurity Threats	5	1	5	- Protect member's access/password carefully - Hardware that is used for the project must be checked carefully.
12	Scalability Issues	2	1	2	- Use resources that can scale

9 Meeting journals

Meeting No: 1

Meeting Details

Date:	09/07/2024
Time:	10:30am – 11:30am
Attendees:	Le Anh Quan (Project team leader)
	Pham Trinh Hoang Long (Project team member)
	Mai Xuan Huy (Project team member)
	Phan Ngoc Quang Anh (Project team member)
	Nguyen Cong Gia Hien (Project team member)
	Mr. Tran Duc Nguyen (TMA supervisor)
	Mr. Tran Nhat Quang (RMIT coordinator)
	Mr. Nguyen Minh Nghia (TMA supervisor)
Apologies:	
Copy To:	

Information /Decision

Item No.	Discussion Summary
1	Project progress update
2	Show website UI (homepage, processing page, result page)
3	Demonstrate the analysis of a video with AI models for vehicle detection and counting within a specified region.
4	Feedback and recommendations from supervisors as well as the coordinator
5	Plan for the next objectives

Action Items

No	Item	Who	By
1	Change the background image of the website, and test responsiveness when running on smartphones. Implement other components of the website.	Web Team	16/07/2024
2	Utilize a video of a location with clearer, cleaner features for analysis to improve demonstration in the future.	AI Team	18/07/2024
3	Research zone-based detection	Whole Team	16/07/2024
4	Document the theoretical grounds of every implementation for demonstration and report writing purposes.	Whole Team	Final deadline

Meeting No: 2

Meeting Details

Date:	16/07/2024
Time:	10:30am – 11:30am
Attendees:	Le Anh Quan (Project team leader)
	Pham Trinh Hoang Long (Project team member)
	Mai Xuan Huy (Project team member)
	Phan Ngoc Quang Anh (Project team member)
	Nguyen Cong Gia Hien (Project team member)
	Mr. Tran Duc Nguyen (TMA supervisor)
	Mr. Nguyen Minh Nghia (TMA supervisor)
Apologies:	Mr. Tran Nhat Quang (RMIT coordinator)
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Information /Decision

Item No.	Discussion Summary
1	Project progress update
2	Demonstrate the analysis of a video with AI models for vehicle detection and counting within a specified region.
3	Feedback and recommendations from supervisors as well as the coordinator
4	Plan for the next objectives

Action Items

No	Item	Who	By
1	Change the background image of the website, and test responsiveness when running on smartphones. Implement other components of the website - Completed.	Web Team	16/07/2024
2	Utilize a video of a location with clearer, cleaner features for analysis to improve demonstration in the future - Completed.	AI Team	18/07/2024
3	Research zone-based detection – Completed	Whole Team	16/07/2024
4	Improve the accuracy of AI models for zone-based counting	AI Team	23/07/2024
5	Implement zone drawing functionality for the website	Web Team	30/07/2024
6	Research object tracking algorithms	Whole Team	23/07/2024

Meeting No: 3

Meeting Details

Date:	23/07/2024
Time:	10:30am – 11:30am
Attendees:	Le Anh Quan (Project team leader)
	Pham Trinh Hoang Long (Project team member)
	Mai Xuan Huy (Project team member)
	Phan Ngoc Quang Anh (Project team member)
	Nguyen Cong Gia Hien (Project team member)
	Mr. Tran Duc Nguyen (TMA supervisor)
	Mr. Nguyen Minh Nghia (TMA supervisor)
Apologies:	Mr. Tran Nhat Quang (RMIT coordinator)
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Information /Decision

Item No.	Discussion Summary
1	Project progress update
2	Feedback and recommendations from supervisors as well as the coordinator
3	Plan for the next objectives

Action Items

No	Item	Who	By
1	Improve accuracy of AI models for zone-based counting - Completed	AI Team	23/07/2024
2	Implement zone drawing functionality for the website – In progress	Web Team	30/07/2024
3	Research object tracking algorithms - Completed	Whole team	23/07/2024
4	Train street sign detection models	AI Team	30/07/2024

5	Train vehicle direction tracking model	AI Team	30/07/2024
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Meeting No: 4

Meeting Details

Date:	30/07/2024
Time:	10:30am – 11:30am
Attendees:	Le Anh Quan (Project team leader)
	Pham Trinh Hoang Long (Project team member)
	Mai Xuan Huy (Project team member)
	Phan Ngoc Quang Anh (Project team member)
	Nguyen Cong Gia Hien (Project team member)
	Mr. Tran Duc Nguyen (TMA supervisor)
	Mr. Tran Nhat Quang (RMIT coordinator)
	Mr. Nguyen Minh Nghia (TMA supervisor)
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Information /Decision

Item No.	Discussion Summary
1	Project progress update
2	Show website UI (loading screen, detection zone drawing window)
3	Demonstrate AI models for street sign detection
4	Demonstrate the AI models for vehicle direction tracking.
5	Feedback and recommendations from supervisors as well as the coordinator
6	Plan for the next objectives

Action Items

No	Item	Who	By
1	Implement zone drawing functionality for the website - Completed	Web Team	30/07/2024
2	Train street sign detection models - Completed	AI Team	30/07/2024
3	Train vehicle direction tracking model - Completed	AI Team	30/07/2024
4	Research the approach of shape-drawing by enlarging a provided shape for the desired zone.	Web Team	06/08/2024
5	Fix detection area drawing bugs, and improve visibility of annotations on the zone drawing modal.	Web Team	06/08/2024
6	Research databases for the system	Whole Team	06/08/2024
7	Record drawbacks or limitations for documentation	Whole Team	Final deadline

Meeting No: 5

Meeting Details

Date:	06/08/2024
Time:	10:30am – 11:30am
Attendees:	Le Anh Quan (Project team leader)
	Pham Trinh Hoang Long (Project team member)
	Mai Xuan Huy (Project team member)
	Phan Ngoc Quang Anh (Project team member)
	Nguyen Cong Gia Hien (Project team member)
	Mr. Tran Duc Nguyen (TMA supervisor)
	Mr. Tran Nhat Quang (RMIT coordinator)
	Mr. Nguyen Minh Nghia (TMA supervisor)
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Information /Decision

Item No.	Discussion Summary
1	Project progress update, confirm important timelines
2	Show website UI (updated detection zone drawing window)
3	Demonstrate database research approaches (SQLite, PostgreSQL)
4	Demonstrate improvements in the AI models.
5	Feedback and recommendations from supervisors as well as the coordinator
6	Plan for the next objectives

Action Items

No	Item	Who	By
1	Research the approach of shape-drawing by enlarging a provided shape for the desired zone - Completed.	Web Team	06/08/2024
2	Fix detection area drawing bugs, and improve visibility of annotations on the zone drawing modal - Completed.	Web Team	06/08/2024
3	Research databases for the system - Completed	Whole Team	06/08/2024
4	Find methods to ensure the coordinates of the zone drawn by the user are correctly used by the AI models.	Whole Team	13/08/2024
5	Choose to prioritize the functionalities or the accuracy of the application based on current progress.	Whole Team	13/08/2024
6	Use MinIO as recommended by supervisors for the database	Whole Team	13/08/2024
7	Utilize metrics available with the AI models to evaluate their performance.	AI Team	13/08/2024

*Meeting No: 6***Meeting Details**

Date:	13/08/2024
Time:	10:30am – 11:30am
Attendees:	Le Anh Quan (Project team leader)
	Pham Trinh Hoang Long (Project team member)
	Mai Xuan Huy (Project team member)
	Phan Ngoc Quang Anh (Project team member)
	Nguyen Cong Gia Hien (Project team member)
	Mr. Tran Duc Nguyen (TMA supervisor)
	Mr. Tran Nhat Quang (RMIT coordinator)
	Mr. Nguyen Minh Nghia (TMA supervisor)
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Information /Decision

Item No.	Discussion Summary
1	Project progress update, confirm important timelines
2	Show website UI
3	Demonstrate API endpoints of the MinIO database and test operations.
4	Demonstrate improvements in the AI models.
5	Feedback and recommendations from supervisors as well as the coordinator
6	Plan for the next objectives

Action Items

No	Item	Who	By
1	Find methods to ensure coordinates of the zone drawn by the user are correctly used by the AI models - Completed.	Whole Team	13/08/2024
2	Use MinIO as recommended by supervisors for the database - Completed	Whole Team	13/08/2024
3	Utilize metrics available with the AI models to evaluate their performance - Completed.	AI Team	13/08/2024
4	Choose to prioritize the functionalities or the accuracy of the application based on current progress - Completed.	Whole Team	13/08/2024
5	Implement a 10-point zone drawing to ensure coverage of most scenarios	Web Team	27/08/2024
6	Fix detection area drawing operation for better visualization of the regions.	Web Team	20/08/2024
7	Add a progress indicator for loading screen	Web Team	20/08/2024
8	Apply desirable images' resizing before insert into AI models	AI Team	20/08/2024

Meeting No: 7

Meeting Details

Date:	20/08/2024
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Time:	10:30am – 11:30am
Attendees:	Le Anh Quan (Project team leader)
	Pham Trinh Hoang Long (Project team member)
	Mai Xuan Huy (Project team member)
	Phan Ngoc Quang Anh (Project team member)
	Nguyen Cong Gia Hien (Project team member)
	Mr. Tran Duc Nguyen (TMA supervisor)
	Mr. Tran Nhat Quang (RMIT coordinator)
	Mr. Nguyen Minh Nghia (TMA supervisor)
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Information /Decision

Item No.	Discussion Summary
1	Project progress update
2	Show entire flow of the system
3	Demonstrate improvements of the AI models (plate recognition)
4	Plan for next objectives

Action Items

No	Item	Who	By
1	Implement 10-point zone drawing to ensure coverage of most scenarios - In progress	Web Team	27/08/2024
2	Fix detection area drawing operation for better visualization of the regions - Completed	Web Team	20/08/2024
3	Add progress indicator for loading screen – Delayed due to low priority	Web Team	20/08/2024
4	Apply desirable images' resizing before insert into AI models - Completed	AI Team	20/08/202
5	Prepare for project showcase event	Whole Team	05/09/2024
6	Record duration of each step of the analysis with AI to identify bugs	AI Team	27/08/2024
7	Research PaddleOCR for plate recognition	Web Team	27/08/2024

Meeting No: 8

Meeting Details

Date:	27/08/2024
Time:	10:30am – 11:30am
Attendees:	Le Anh Quan (Project team leader)
	Pham Trinh Hoang Long (Project team member)
	Mai Xuan Huy (Project team member)
	Phan Ngoc Quang Anh (Project team member)
	Nguyen Cong Gia Hien (Project team member)

	Mr. Tran Duc Nguyen (TMA supervisor)
	Mr. Tran Nhat Quang (RMIT coordinator)
	Mr. Nguyen Minh Nghia (TMA supervisor)
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Information /Decision

Item No.	Discussion Summary
1	Project progress update
2	Plan for next objectives

Action Items

No	Item	Who	By
1	Prepare for project showcase event – In progress	Whole Team	13/09/2024
2	Record duration of each step of the analysis with AI to identify bugs - Completed	AI Team	27/08/2024
3	Research PaddleOCR for plate recognition - Completed	Web Team	27/08/2024

Meeting No: 9

Meeting Details

Date:	05/09/2024
Time:	10:30am – 11:30am
Attendees:	Le Anh Quan (Project team leader)
	Pham Trinh Hoang Long (Project team member)
	Mai Xuan Huy (Project team member)
	Phan Ngoc Quang Anh (Project team member)
	Nguyen Cong Gia Hien (Project team member)
	Mr. Tran Duc Nguyen (TMA supervisor)
	Mr. Tran Nhat Quang (RMIT coordinator)
	Mr. Nguyen Minh Nghia (TMA supervisor)
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Information /Decision

Item No.	Discussion Summary
1	Project progress update, confirm important timelines
2	Show evaluation of analysis time running on different GPUs
3	Inform existing bug with zone drawing function which results in inaccurate zone coordinates
4	Discuss probability of completing parking violation analysis feature with given timeline
5	Feedback and recommendations from supervisors as well as coordinator

6	Plan for next objectives
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Action Items

No	Item	Who	By
1	Prepare for project showcase event – In progress	Whole Team	10/09/2024
2	Fix zone drawing bug for correctly dimensioned zoned	Web Team	10/09/2024
3	Integrate the plate recognition module to the pipeline	Whole Team	17/09/2024
4	Research enhancement methods for possible future development: Quanty, pruning, ...	Web Team	10/09/2024
5	Analyze shorter videos for demonstration at showcase event	AI Team	10/09/2024

Meeting No: 10

Meeting Details

Date:	10/09/2024
Time:	10:30am – 11:30am
Attendees:	Le Anh Quan (Project team leader)
	Mai Xuan Huy (Project team member)
	Phan Ngoc Quang Anh (Project team member)
	Nguyen Cong Gia Hien (Project team member)
	Mr. Tran Duc Nguyen (TMA supervisor)
	Mr. Tran Nhat Quang (RMIT coordinator)
	Mr. Nguyen Minh Nghia (TMA supervisor)
Apologies:	Pham Trinh Hoang Long (Project team member)
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Information /Decision

Item No.	Discussion Summary
1	Project progress update, confirm important timelines
2	Feedback and recommendations from supervisors as well as coordinator
3	Plan for next objectives

Action Items

No	Item	Who	By
1	Prepare for project showcase event – Completed	Whole Team	10/09/2024
2	Fix zone drawing bug for correctly dimensioned zoned - Completed	Web Team	10/09/2024
3	Integrate the plate recognition module to the pipeline – In progress	Whole Team	17/09/2024
4	Research enhancement methods for possible future development: Quanty, pruning, ... - Completed	Web Team	10/09/2024
5	Analyze shorter videos for demonstration at showcase event - Completed	AI Team	10/09/2024

6	Finish Final Report	Whole Team	17/09/2024
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